REPORT

APPLICATION OF AI SEARCH AND OPTIMIZATION ALGORITHMS IN REAL-WORLD SCENARIOS

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Course: Artificial Intelligence

Topic:

Scenario-Based Analysis of AI Search, Optimization, CSP, and Game AI Algorithms

ABSTRACT

Artificial Intelligence algorithms play a significant role in solving complex real-world problems that involve uncertainty, large search spaces, and dynamic environments. This report analyzes five different AI-based scenarios including emergency drone navigation, smart traffic optimization, autonomous rover exploration, exam timetable generation, and strategic game decision-making. Various AI techniques such as Greedy Best-First Search, A*, Hill Climbing, Simulated Annealing, Genetic Algorithms, Online Search, Constraint Satisfaction Problems, Minimax, and Alpha-Beta Pruning are studied. Each algorithm is evaluated based on efficiency, adaptability, optimality, scalability, and suitability for real-time applications. The report identifies the most appropriate algorithm for each scenario with proper justification.

1. INTRODUCTION

Artificial Intelligence enables machines to make intelligent decisions by searching through possible solutions, optimizing performance, and adapting to changing environments. Many real-world problems cannot be solved using traditional methods due to uncertainty, complexity, and continuous changes.

Search algorithms help AI systems find optimal solutions by exploring possible states, while optimization algorithms improve solutions by reducing cost and increasing efficiency. Constraint-based approaches solve problems with multiple restrictions, and game algorithms support intelligent decision-making.

This report focuses on analyzing different AI algorithms based on real-world applications and selecting the most suitable approaches.

2. AI-BASED DRONE NAVIGATION FOR EMERGENCY MEDICINE DELIVERY

2.1 Scenario Description

A government deploys AI-powered drones to deliver emergency medicines in flood-affected regions. The drone environment is highly dynamic because roads may become blocked, weather conditions may change, and travelling costs may vary. The drone uses heuristic information such as straight-line distance and partial real-time updates to select safe and fast routes.

2.2 Greedy Best-First Search Analysis

Greedy Best-First Search selects the node that appears closest to the destination based only on heuristic estimation.

Evaluation Function:

f(n)=h(n)f(n)=h(n)f(n)=h(n)

where,

h(n) = estimated distance from current position to destination.

Working:

The drone immediately selects the path with the smallest estimated distance without considering actual travelling cost.

Advantages:

- Provides faster decisions
- Requires less memory
- Suitable for emergency situations

Limitations:

- May select unsafe routes
- Does not consider weather and terrain cost
- May produce non-optimal solutions

2.3 A* Algorithm Analysis

A* combines actual cost and heuristic estimation.

Evaluation Function:

f(n)=g(n)+h(n)f(n)=g(n)+h(n)f(n)=g(n)+h(n)

where:

- **g(n)** = cost already travelled
- **h(n)** = estimated remaining cost

Working:

A* evaluates both current travel cost and future distance before selecting a route.

Advantages:

- Provides optimal solutions
- Considers terrain and weather conditions
- Improves safety

Limitation:

- Requires more memory and computation

2.4 Impact of Heuristic Accuracy

The accuracy of heuristic information directly affects performance.

- Accurate heuristic improves search speed.
- Poor heuristic increases unnecessary exploration.
- Greedy search is more sensitive because it depends completely on heuristic value.
- A* can still perform efficiently because it considers actual cost.

2.5 Effect of Dynamic Changes

Blocked roads and changing weather conditions can make existing routes invalid.

The system requires:

- Continuous monitoring
- Route recalculation
- Dynamic path planning

2.6 Recommended Algorithm

Dynamic A* Search

A* is recommended because it provides a balance between speed, safety, and optimality. It considers both current cost and future estimation, making it suitable for emergency drone delivery.

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3. SMART CITY TRAFFIC SIGNAL OPTIMIZATION

3.1 Scenario Description

A smart city uses AI to optimize traffic signals at multiple intersections. The objective is to reduce waiting time, fuel consumption, and congestion while continuously adapting to traffic changes.

3.2 Problem Formulation as Optimization Problem

The objective is:

Minimize(Total Waiting Time+Fuel Consumption+Congestion)Minimize(Total\ Waiting\ Time + Fuel\ Consumption + Congestion)Minimize(Total Waiting Time+Fuel Consumption+Congestion)

Variables:

- Signal timing duration
- Green light intervals
- Traffic priorities

Constraints:

- Road capacity
- Safety rules
- Pedestrian movement
- Maximum waiting time

3.3 Hill Climbing Algorithm

Hill Climbing improves the current solution by selecting better neighbouring solutions.

Working:

1. Select current traffic timing.
2. Generate possible changes.
3. Choose the best improvement.
4. Continue until no improvement exists.

Limitations:

- Local maxima problem
- Plateau problem
- Ridge problem

3.4 Advanced Optimization Techniques

Simulated Annealing

Simulated Annealing accepts some worse solutions temporarily to escape local optimum.

Genetic Algorithm

Genetic Algorithm uses selection, crossover, and mutation to generate better solutions.

3.5 Recommended Algorithm

Genetic Algorithm

Genetic Algorithm is recommended because traffic optimization contains a huge search space and continuously changing conditions. It provides scalability and adaptability.

4. AUTONOMOUS MARS ROVER NAVIGATION

4.1 Scenario Description

A Mars rover explores unknown terrain where complete environmental information is unavailable. It must safely collect samples while avoiding hazards.

4.2 Need for Online Search

Offline search requires complete knowledge of the environment, which is unavailable on Mars.

Online search allows the rover to:

- Observe surroundings
- Update knowledge
- Make real-time decisions

4.3 Challenges

- Partial observability
- Unknown obstacles
- Limited sensors
- Communication delays

4.4 Internal Model Development

The rover builds its internal model using:

- Sensor readings
- Terrain information
- Previous experiences

- Hazard detection

The model is updated continuously during exploration.

4.5 Recommended Approach

Online Informed Search

It provides adaptability, safety, and decision-making ability under uncertainty.

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5. AI-BASED EXAM TIMETABLE GENERATION USING CSP

5.1 Scenario Description

A university requires an AI system to generate examination timetables while satisfying multiple academic constraints.

5.2 CSP Formulation

A Constraint Satisfaction Problem contains:

Variables:

Different examinations.

Domains:

Available time slots and examination halls.

Constraints:

- No student conflicts
- Faculty availability
- Hall limitations
- Subject ordering

5.3 Backtracking Search

Backtracking assigns values step-by-step and checks constraints.

If a conflict occurs, the system returns and tries another option.

5.4 Constraint Propagation

Forward checking removes invalid choices before assignment.

Benefits:

- Reduces search space
- Detects conflicts early
- Improves efficiency

5.5 Recommended Strategy

CSP with Backtracking and Forward Checking

This approach provides better scalability and handles complex scheduling constraints.

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6. STRATEGIC GAME AI USING MINIMAX AND ALPHA-BETA PRUNING

6.1 Scenario Description

A gaming company develops an AI opponent capable of making intelligent decisions against human players within limited time.

6.2 Minimax Algorithm

Minimax models two players:

- MAX → AI player
- MIN → Human player

The AI selects the move that maximizes winning chances while considering opponent actions.

6.3 Computational Challenges

Large game trees create:

- High computation cost
- Memory requirements
- Slow decision-making

Complexity:

O(bd)O(b^d)O(bd)

6.4 Alpha-Beta Pruning

Alpha-Beta pruning removes unnecessary branches from the game tree.

Benefits:

- Reduces search time
- Allows deeper exploration
- Maintains optimal decisions

6.5 Evaluation Function

The evaluation function considers:

- Player score
- Resources

- Position advantage
- Defence capability
- Future opportunities

6.6 Recommended Strategy

Minimax + Alpha-Beta Pruning + Heuristic Evaluation

This provides a balance between intelligence, accuracy, and real-time performance.

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7. COMPARATIVE ANALYSIS

Scenario	Algorithm Used	Main Advantage
Emergency Drone Navigation	Dynamic A*	Safe and optimal routing
Traffic Signal Optimization	Genetic Algorithm	Large-scale optimization
Mars Rover Exploration	Online Search	Handles uncertainty
Exam Scheduling	CSP	Constraint management
Game AI	Minimax + Alpha-Beta	Fast decision-making

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8. CONCLUSION

AI algorithms provide effective solutions for complex real-world problems involving uncertainty, optimization, and decision-making. A* search is suitable for emergency navigation because it balances cost and heuristic information. Genetic Algorithms provide better solutions for large optimization problems such as traffic management. Online search enables autonomous systems like Mars rovers to operate safely in unknown environments. CSP techniques efficiently solve scheduling problems with multiple constraints, while Minimax with Alpha-Beta pruning supports intelligent game decision-making. The selection of appropriate AI algorithms depends on factors such as scalability, adaptability, optimality, and real-time requirements.